Research Question

Does renewable energy adoption buffer the impact of oil price fluctuations on economic growth across oil-importing countries over time?

Motivation

Recent geopolitical tensions, most notably the escalating US-Iran conflict, have highlighted the vulnerability of oil-importing nations to sudden disruptions in global energy markets. This project investigates whether renewable energy adoption serves as a structural buffer against oil price shocks, reducing a country's exposure to price volatility and its impact on economic growth. Understanding this relationship is increasingly urgent for policymakers seeking to justify clean energy transitions as a matter of national economic resilience.

Data Sources

The primary dataset is the Global Data on Sustainable Energy (2000–2020) by IEA / World Bank, which provides annual renewable energy share, GDP growth, and GDP per capita for 173 countries. Oil price data is drawn from Brent Crude Oil Prices, with daily prices aggregated to annual averages and year-on-year percentage changes. Energy import dependency, measuring the share of energy consumption sourced from imports, is obtained from the World Bank Open Data portal. All datasets are freely available and downloadable in CSV format.

Data Collection and Characteristics

All datasets are downloaded in CSV format and merged by country and year using Python, producing a panel dataset of 1,473 observations (91 oil-importing countries × 21 years, 2000–2020). Variables include renewable energy share, GDP growth, GDP per capita, energy import dependency, oil price, and annual oil price change. Countries with positive energy import dependency are retained to ensure the analysis focuses exclusively on oil-importing nations.

Planned Analysis

Analysis proceeds from exploratory data analysis and correlation tests to fixed effects panel regression, including an interaction term between oil price change and renewable energy share to test whether renewable energy buffers the economic impact of oil price fluctuations. Machine learning methods including Random Forest, XGBoost, and SHAP values will be applied to assess feature importance. All work is conducted in Python (Jupyter Notebook).